Algebra and Discrete Mathematics
ADM_A_B

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Course Outline

- Vectors and matrices
- System of linear equations
- Matrix inverse and determinants
- Vector spaces, bases and matrix transformations
- Quiz one
- Matrix transformations, fundamental spaces and decompositions
- Eulerian tours and Hamiltonian cycles
- Shortest Paths, Project Scheduling, and Networks
- Quiz two
- Network Flows, Trees, and Matchings
- Make-up Quizzes

Recommended reading

- Saoub, K. R. (2017). A tour through graph theory. Chapman and Hall/CRC.
  - Sections 3.1, 3.2, 4.1, 4.2
  - Accessible online (free copy)
  - Alternative download link

Lecture outline

- Dijkstra's Algorithm
- Chinese Postman Problem
- Project scheduling
- Critical path
- Trees
- Spanning trees
- Shortest networks
- Steiner Trees
- Metric Traveling Salesman Problem

Shortest Paths, Project Scheduling, and Networks

- Dijkstra's Algorithm
- Chinese Postman Problem
- Project scheduling
- Critical path
- Trees
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Shortest path problem

- A path is a sequence of vertices in which there is an edge between consecutive vertices and no vertex is repeated
- Weight: distance, cost, time, etc.
- Shortest path: the path of the least total weight
- A shortest path between any two vertices exists if the graph is connected
- Scenario: fastest route to travel from one location to another

Dijkstra's Algorithm

- Proposed in 1956 by Edsger W. Dijkstra.
- Almost every GIS (Geographic Information System, or mapping software) uses a modification of Dijkstra's algorithm to provide directions
- Numerous versions of Dijkstra's Algorithm exist
- See original algorithm: DIJKSTRA, E. (1959). A Note on Two Problems in Connexion with Graphs. Numerische Mathematik, 1, 269-271.

Dijkstra's Algorithm – notations

- Each vertex is given a two-part label

$$L(v) = (x, \omega(v))$$

- x: the predecessor of v on the current best known path
- $\omega(v)$: the weight of the path that was used to get to v from the designated starting vertex
- F: a set of vertices that are not highlighted yet

Dijkstra's Algorithm – input and output

- **Input**: Weighted connected graph $G = (V, E)$ with nonnegative edge weights and vertices designated as *Start* and *End*
- **Output**: A shortest path from *Start* to *End* and its total weight $\omega(\mathit{End})$.

Dijkstra's Algorithm – steps

1. For each vertex v of G, assign a label $L(v)$:

$$L(v) = \begin{cases} (-, 0), & \text{if } v \text{ is } \textit{Start} \\ (-, \infty), & \text{Otherwise} \end{cases}$$

   Highlight *Start*. $F = V - \{\textit{Start}\}$. Let the *current vertex* be *Start*.

2. Update the labels for each vertex v in F that is a neighbor of *current vertex*, say u:

$$L(v) = \begin{cases} (u, \omega(u) + \omega(uv)), & \text{if } \omega(u) + \omega(uv) < \omega(v) \\ L(v), & \text{Otherwise} \end{cases}$$

3. Highlight a vertex v in F with the lowest weight and the edge indicated by its predecessor label. Remove v from F. Redefine the *current vertex* to be v.

4. Repeat Steps 2 and 3 until the vertex *End* has been highlighted.

Dijkstra's Algorithm – steps

5. The shortest path from *Start* to *End* is found by tracing back from *End* using the first component of the labels. The total weight of the path is the weight for *End* given in the second component of its label.

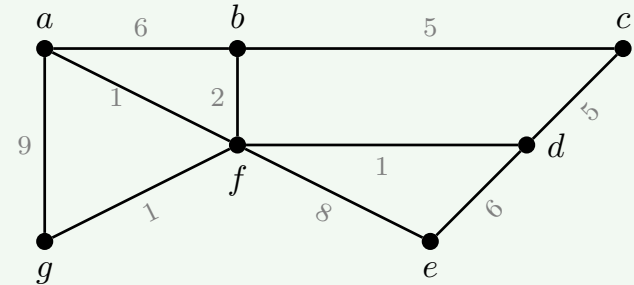
Note

- The set F of vertices consists of all un-highlighted vertices and all are under consideration for becoming the next highlighted vertex
- It is important that we do not only consider the neighbors of the last vertex highlighted, as a path from a previously chosen vertex may in fact lead to the shortest path

Dijkstra's Algorithm – example

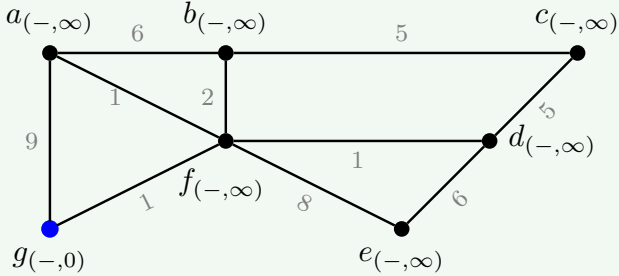
Example

Take *Start* $= g$ and *End* $= c$.



Dijkstra's Algorithm – example

Example



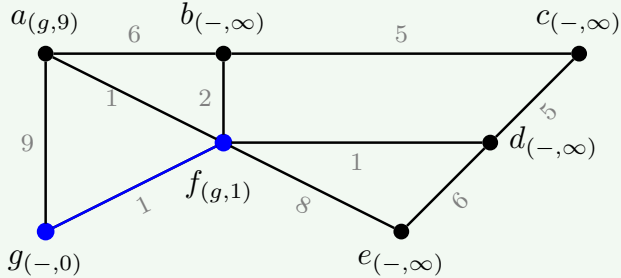
- Step 1. Highlight g. Label

$$L(v) = \begin{cases} (-,0) & v = g \\ (-,\infty) & \text{Otherwise} \end{cases}$$

$F = \{a, b, c, d, e, f\}$. The *current vertex* is g.

Dijkstra's Algorithm – example

Example



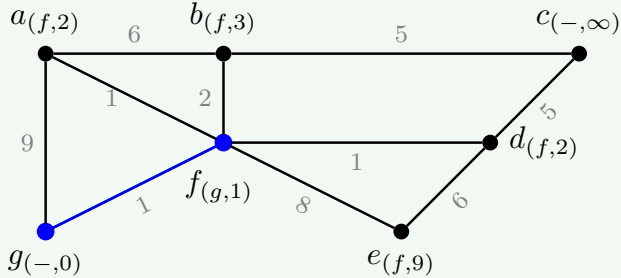
- Step 2. $F = \{a, b, c, d, e, f\}$. The neighbors of g are a and f.

$$\begin{aligned} \omega(g) + \omega(ga) &= 0 + 9 = 9 < \infty = \omega(a) \Longrightarrow L(a) = (g, 9) \\ \omega(g) + \omega(gf) &= 0 + 1 = 1 < \infty = \omega(f) \Longrightarrow L(f) = (g, 1) \end{aligned}$$

- Step 3. minimum weight: f; highlight gf and f. $F = \{a, b, c, d, e\}$. *current vertex* is f.

Dijkstra's Algorithm – example

Example

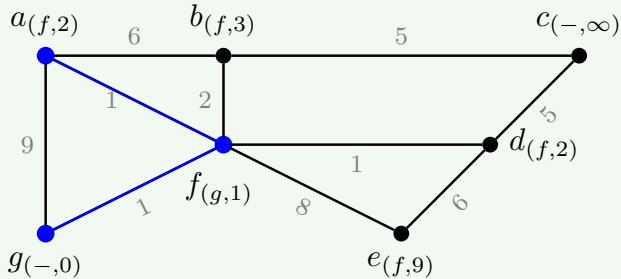


- Step 2. $F = \{a, b, c, d, e\}$. Neighbors of f: a, b, d, e

$$
\begin{aligned}
\omega(f) + \omega(fa) &= 1 + 1 = 2 < 9 = \omega(a) \Longrightarrow L(a) = (f, 2) \\
\omega(f) + \omega(fb) &= 1 + 2 = 3 < \infty = \omega(b) \Longrightarrow L(b) = (f, 3) \\
\omega(f) + \omega(fd) &= 1 + 1 = 2 < \infty = \omega(d) \Longrightarrow L(d) = (f, 2) \\
\omega(f) + \omega(fe) &= 1 + 8 = 9 < \infty = \omega(e) \Longrightarrow L(e) = (f, 9)
\end{aligned}
$$

Dijkstra's Algorithm – example

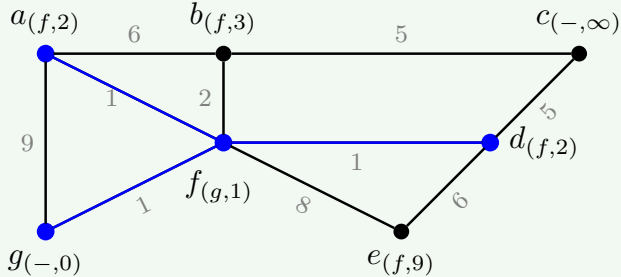
Example



- Step 3. $F = \{a, b, c, d, e\}$. The minimum weight among the vertices in F is attained by both a and d. Choose one of them arbitrarily.
  - Let us highlight fa and a.
  - $F = \{b, c, d, e\}$.
  - *current vertex* is a.

Dijkstra's Algorithm – example

Example



- Step 2. $F = \{b, c, d, e\}$. Neighbor of a: b

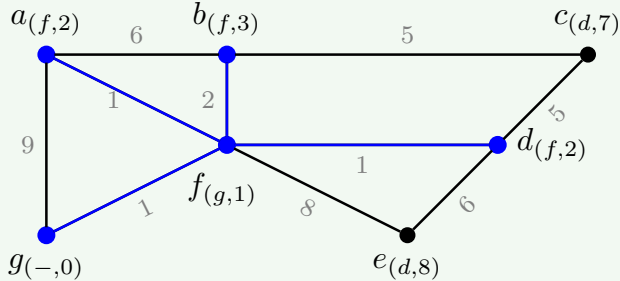
$$\omega(a) + \omega(ba) \quad = \quad 2 + 6 = 8 > 3 = \omega(b)$$

  We do not update the label of b.

- Step 3. Highlight fd and d. $F = \{b, c, e\}$. *current vertex* is d.

Dijkstra's Algorithm – example

Example



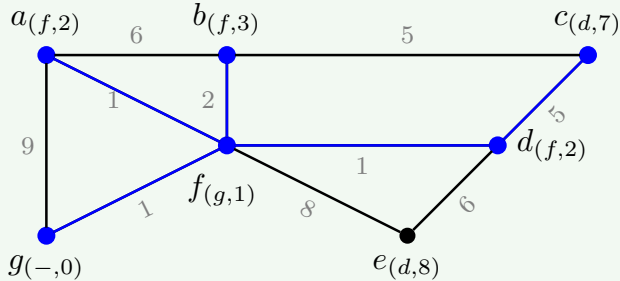
- Step 2. $F = \{b, c, e\}$. Neighbors of d: c, e

$$\begin{aligned} \omega(d) + \omega(dc) &= 2 + 5 = 7 < \infty = \omega(c) \Longrightarrow L(c) = (d, 7) \\ \omega(d) + \omega(de) &= 2 + 6 = 8 < 9 = \omega(e) \Longrightarrow L(e) = (d, 8) \end{aligned}$$

- Step 3. Highlight fb and b. $F = \{c, e\}$. *current vertex* is b.

Dijkstra's Algorithm – example

Example



- Step 2. $F = \{c, e\}$. Neighbor of b: $c\ \omega(b) + \omega(bc) = 3 + 5 = 8 > 7 = \omega(c)$
- Step 3. Highlight dc and c.
- Step 4. This terminates the iterations since we have reached *End*
- Step 5. The shortest path from g to c is $g \to f \to d \to c$, with total weight 7.

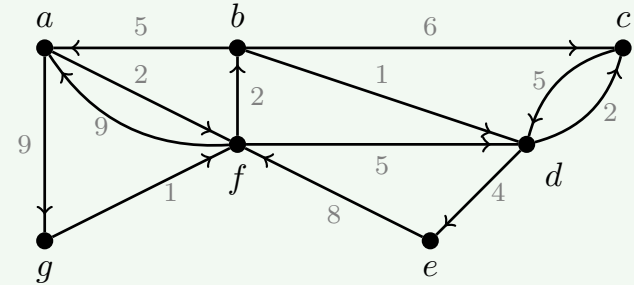
Dijkstra's Algorithm for digraphs

- A digraph is a graph in which the edges now have a direction associated to them, which could be used to model a one-way street.
- For an arc yx, x is the *head* and y is the *tail*.
- In Step 2, instead of all neighbors, we consider only the *out-neighbors* of the current vertex, that is, vertices joined to it by outgoing arcs.

Dijkstra's Algorithm for digraphs – example

Example

Start= g, *End*= c



We record the changes in the following table.

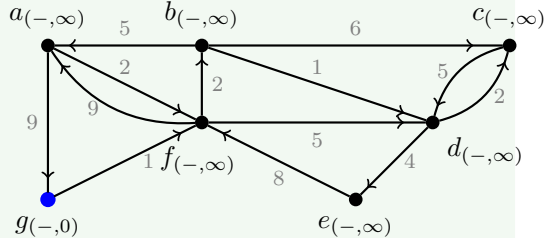
g	a	b	c	d	e	f

Dijkstra's Algorithm for digraphs – example

Example

- Step 1. *current vertex* is g,
  $F = \{a, b, c, d, e, f\}$

g	a	b	c	d	e	f
0	∞	∞	∞	∞	∞	∞



Dijkstra's Algorithm for digraphs – example

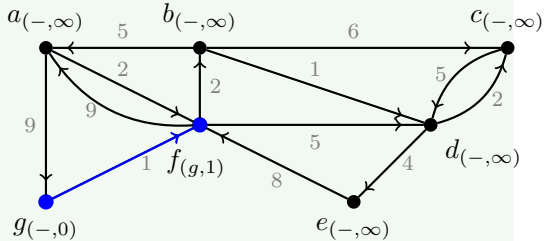
Example

- Step 2. *current vertex* is g,
  $F = \{a, b, c, d, e, f\}$,
  out-neighbors of g: f

$$\omega(g) + \omega(gf) = 0 + 1 < \infty$$

- Step 3. *current vertex* is f,
  $F = \{a, b, c, d, e\}$

	a	b	c	d	e	f
	∞	∞	∞	∞	∞	∞
$g(0)$	∞	∞	∞	∞	∞	$(g, 1)$

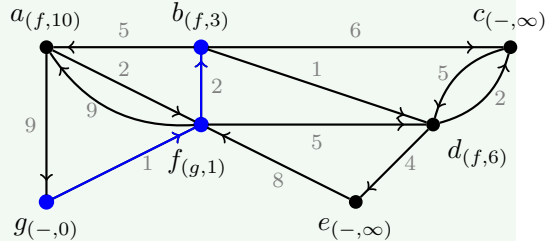


Dijkstra's Algorithm for digraphs – example

Example

- Step 2. *current vertex* is f,
  $F = \{a, b, c, d, e\}$,
  out-neighbors of f: a, b, d

$$\omega(f) + \omega(fa) = 1 + 9 = 10 < \infty,$$
$$\omega(f) + \omega(fb) = 1 + 2 = 3 < \infty,$$
$$\omega(f) + \omega(fd) = 1 + 5 = 6 < \infty.$$



- Step 3. *current vertex* is b,
  $F = \{a, c, d, e\}$

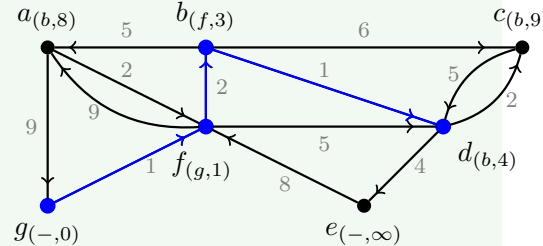
	a	b	c	d	e	f
	∞	∞	∞	∞	∞	∞
$g(0)$	∞	∞	∞	∞	∞	$(g,1)$
$f(1)$	$(f,10)$	$(f,3)$	∞	$(f,6)$	∞	$(g,1)$

Dijkstra's Algorithm for digraphs – example

Example

- Step 2. *current vertex* is b,
  $F = \{a, c, d, e\}$,
  out-neighbors of b: a, c, d

$$\omega(b) + \omega(ba) = 3 + 5 = 8 < 10,$$
$$\omega(b) + \omega(bc) = 3 + 6 = 9 < \infty,$$
$$\omega(b) + \omega(bd) = 3 + 1 = 4 < 6.$$



- Step 3. *current vertex* is d, $F = \{a, c, e\}$

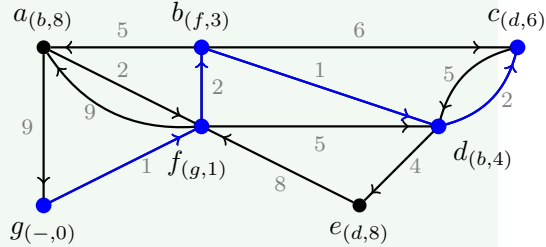
	a	b	c	d	e	f
	∞	∞	∞	∞	∞	∞
$g(0)$	∞	∞	∞	∞	∞	$(g, 1)$
$f(1)$	$(f, 10)$	$(f, 3)$	∞	$(f, 6)$	∞	$(g, 1)$
$b(3)$	$(b, 8)$	$(f, 3)$	$(b, 9)$	$(b, 4)$	∞	$(g, 1)$

Dijkstra's Algorithm for digraphs – example

Example

- Step 2. *current vertex* is d, $F = \{a, c, e\}$, out-neighbors of d: c, e

$$\omega(d) + \omega(dc) = 4 + 2 = 6 < 9,$$
$$\omega(d) + \omega(de) = 4 + 4 = 8 < \infty$$



- Step 3. *current vertex* is c
- Step 4. we have reached *End*
- Step 5. $g \to f \to b \to d \to c$, weight 6

	a	b	c	d	e	f
	∞	∞	∞	∞	∞	∞
$g(0)$	∞	∞	∞	∞	∞	$(g,1)$
$f(1)$	$(f,10)$	$(f,3)$	∞	$(f,6)$	∞	$(g,1)$
$b(3)$	$(b,8)$	$(f,3)$	$(b,9)$	$(b,4)$	∞	$(g,1)$
$d(4)$	$(b,8)$	$(f,3)$	$(d,6)$	$(b,4)$	$(d,8)$	$(g,1)$

Remark

- It is possible for a path not to exist from one vertex to another based upon the direction of the arcs
- For example, if a is the head of every arc incident to a, then no directed path of positive length can start at a.
- For digraphs, if all unhighlighted vertices have weight ∞ before *End* is reached, the algorithm terminates (Step 3) and reports that no directed path from *Start* to *End* exists.

Shortest Paths, Project Scheduling, and Networks

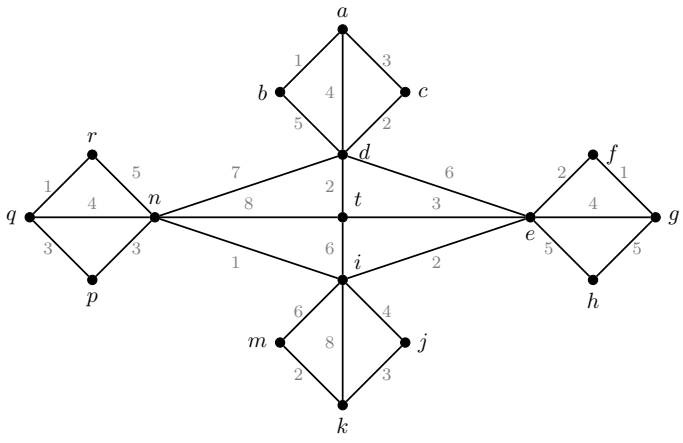
- Dijkstra's Algorithm
- Chinese Postman Problem
- Project scheduling
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Chinese Postman Problem

- Recall: for a connected, undirected graph with nonnegative edge weights, the *Chinese Postman Problem* is to find a closed walk of minimum total weight that traverses every edge at least once.
- First, Eulerize the graph by pairing odd-degree vertices and duplicating shortest paths between them, minimizing the total added weight.
- Then, find an Eulerian circuit in the resulting graph.
- We now use Dijkstra's Algorithm to find the shortest paths needed for an optimal Eulerization.

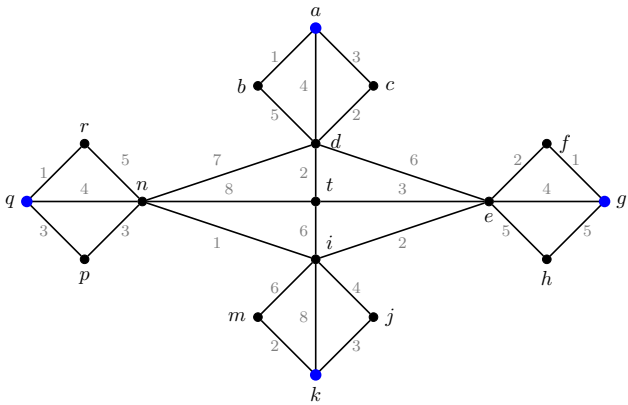
Chinese Postman Problem – example

Use Dijkstra's Algorithm to determine the best pairing of the odd vertices and the total weight of the duplicated edges in an Eulerization of the graph.



Chinese Postman Problem – example

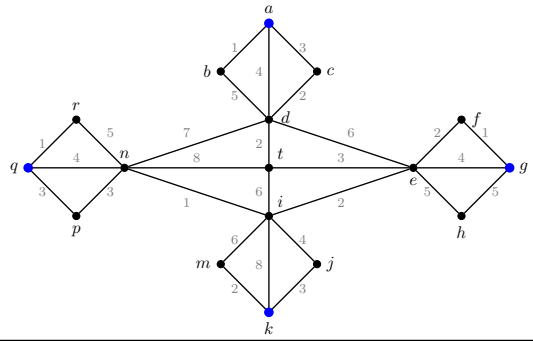
- Four odd vertices: a, g, k, q
- Three possible pairings $a-g, k-q$, or $a-k, g-q$, or $a-q, g-k$



Chinese Postman Problem – example

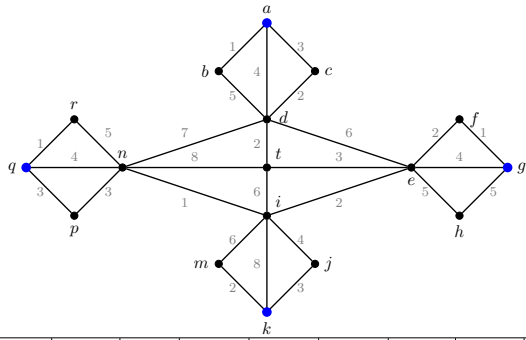
- Three possible pairings $a-g, k-q$, or $a-k, g-q$, or $a-q, g-k$
- Applying Dijkstra's Algorithm, we find the shortest path for each paired set of vertices, and hence the total weight of the duplicated paths needed to Eulerize the graph.

Chinese Postman Problem – example



	b	c	d	n	e	t	r	q	p	f	g	h	i	m	j	k
	∞	∞	∞	∞	∞	∞	∞	∞	∞	∞	∞	∞	∞	∞	∞	∞
$a(0)$	$(a,1)$	$(a,3)$	$(a,4)$	∞	∞	∞	∞	∞	∞	∞	∞	∞	∞	∞	∞	∞
$b(1)$	$(a,1)$	$(a,3)$	$(a,4)$	∞	∞	∞	∞	∞	∞	∞	∞	∞	∞	∞	∞	∞
$c(3)$	$(a,1)$	$(a,3)$	$(a,4)$	∞	∞	∞	∞	∞	∞	∞	∞	∞	∞	∞	∞	∞

Chinese Postman Problem – example



	b	c	d	n	e	t	r	q	p	f	g	h	i	m	j	k
	∞	∞	∞	∞	∞	∞	∞	∞	∞	∞	∞	∞	∞	∞	∞	∞
$a(0)$	$(a,1)$	$(a,3)$	$(a,4)$	∞	∞	∞	∞	∞	∞	∞	∞	∞	∞	∞	∞	∞
$b(1)$	$(a,1)$	$(a,3)$	$(a,4)$	∞	∞	∞	∞	∞	∞	∞	∞	∞	∞	∞	∞	∞
$c(3)$	$(a,1)$	$(a,3)$	$(a,4)$	∞	∞	∞	∞	∞	∞	∞	∞	∞	∞	∞	∞	∞
$d(4)$	$(a,1)$	$(a,3)$	$(a,4)$	$(d,11)$	$(d,10)$	$(d,6)$	∞	∞	∞	∞	∞	∞	∞	∞	∞	∞
$t(6)$	$(a,1)$	$(a,3)$	$(a,4)$	$(d,11)$	$(t,9)$	$(d,6)$	∞	∞	∞	∞	∞	∞	$(t,12)$	∞	∞	∞
$e(9)$	$(a,1)$	$(a,3)$	$(a,4)$	$(d,11)$	$(t,9)$	$(d,6)$	∞	∞	∞	$(e,11)$	$(e,13)$	$(e,14)$	$(e,11)$	∞	∞	∞
$f(11)$	$(a,1)$	$(a,3)$	$(a,4)$	$(d,11)$	$(t,9)$	$(d,6)$	∞	∞	∞	$(e,11)$	$(f,12)$	$(e,14)$	$(e,11)$	∞	∞	∞
$i(11)$	$(a,1)$	$(a,3)$	$(a,4)$	$(d,11)$	$(t,9)$	$(d,6)$	∞	∞	∞	$(e,11)$	$(f,12)$	$(e,14)$	$(e,11)$	$(i,17)$	$(i,15)$	$(i,19)$
$n(11)$	$(a,1)$	$(a,3)$	$(a,4)$	$(d,11)$	$(t,9)$	$(d,6)$	$(n,16)$	$(n,15)$	$(n,14)$	$(e,11)$	$(f,12)$	$(e,14)$	$(e,11)$	$(i,17)$	$(i,15)$	$(i,19)$

Chinese Postman Problem – example

	b	c	d	n	e	t	r	q	p	f	g	h	i	m	j	k
	∞	∞	∞	∞	∞	∞	∞	∞	∞	∞	∞	∞	∞	∞	∞	∞
$a(0)$	$(a,1)$	$(a,3)$	$(a,4)$	∞	∞	∞	∞	∞	∞	∞	∞	∞	∞	∞	∞	∞
$b(1)$	$(a,1)$	$(a,3)$	$(a,4)$	∞	∞	∞	∞	∞	∞	∞	∞	∞	∞	∞	∞	∞
$c(3)$	$(a,1)$	$(a,3)$	$(a,4)$	∞	∞	∞	∞	∞	∞	∞	∞	∞	∞	∞	∞	∞
$d(4)$	$(a,1)$	$(a,3)$	$(a,4)$	$(d,11)$	$(d,10)$	$(d,6)$	∞	∞	∞	∞	∞	∞	∞	∞	∞	∞
$t(6)$	$(a,1)$	$(a,3)$	$(a,4)$	$(d,11)$	$(t,9)$	$(d,6)$	∞	∞	∞	∞	∞	∞	$(t,12)$	∞	∞	∞
$e(9)$	$(a,1)$	$(a,3)$	$(a,4)$	$(d,11)$	$(t,9)$	$(d,6)$	∞	∞	∞	$(e,11)$	$(e,13)$	$(e,14)$	$(e,11)$	∞	∞	∞
$f(11)$	$(a,1)$	$(a,3)$	$(a,4)$	$(d,11)$	$(t,9)$	$(d,6)$	∞	∞	∞	$(e,11)$	$(f,12)$	$(e,14)$	$(e,11)$	∞	∞	∞
$i(11)$	$(a,1)$	$(a,3)$	$(a,4)$	$(d,11)$	$(t,9)$	$(d,6)$	∞	∞	∞	$(e,11)$	$(f,12)$	$(e,14)$	$(e,11)$	$(i,17)$	$(i,15)$	$(i,19)$
$n(11)$	$(a,1)$	$(a,3)$	$(a,4)$	$(d,11)$	$(t,9)$	$(d,6)$	$(n,16)$	$(n,15)$	$(n,14)$	$(e,11)$	$(f,12)$	$(e,14)$	$(e,11)$	$(i,17)$	$(i,15)$	$(i,19)$

$$a \to d \to t \to e \to f \to g$$

with total weight 12

Chinese Postman Problem – example

- Three possible pairings $a-g, k-q$, or $a-k, g-q$, or $a-q, g-k$
- Applying Dijkstra's Algorithm, we find the shortest path for each paired set of vertices, and hence the total weight of the duplicated paths needed to Eulerize the graph.

Path Pairs	**Weight**	**Total Weight**
$a \to d \to t \to e \to f \to g$	12	
$k \to j \to i \to n \to q$	12	24
$a \to d \to t \to e \to i \to j \to k$	18	
$g \to f \to e \to i \to n \to q$	10	28
$a \to d \to n \to q$	15	
$g \to f \to e \to i \to j \to k$	12	27

Exercise: verify these shortest paths and total weights.

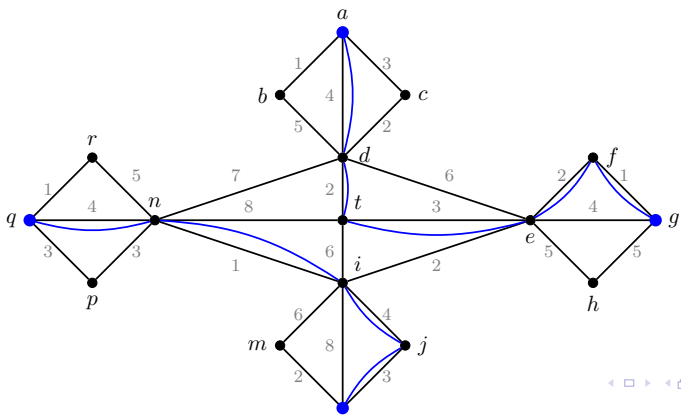
Chinese Postman Problem – example

- Once the shortest paths have been found, we duplicate the edges along those paths to obtain an Eulerization.
- However, the paths in a pairing may share edges.
- For example, both paths in the second pairing contain the edge ei.
- We can omit both copies of ei and retain the other duplicated edges.
- Removing both copies preserves the parity of every vertex degree and reduces the total added weight by $2\omega\,(ei) = 4$, making this Eulerization equal in cost to the first one.

$adteijk$	18	
$gfeinq$	10	28

Chinese Postman Problem – example

- duplicate the edges along $a \to d \to t \to e \to f \to g$ and $k \to j \to i \to n \to q$.
- duplicate the edges along $a \to d \to t \to e \to i \to j \to k$ and $g \to f \to e \to i \to n \to q$, except for the shared edge ei, which is omitted from both paths.
- Both choices give an Eulerization with total added weight 24.



Remarks

- The use of Dijkstra's Algorithm allows for a methodical approach to the Chinese Postman Problem.
- However, even for small examples, the number of shortest paths that must be considered can be quite large.
- A more efficient approach uses matchings.

Shortest Paths, Project Scheduling, and Networks

- Dijkstra's Algorithm
- Chinese Postman Problem
- Project scheduling
- Critical path
- Trees
- Spanning trees
- Shortest networks
- Steiner Trees
- Metric Traveling Salesman Problem

Definitions

Definition

Consider a project consisting of multiple tasks.

- *Task*: a required step of a project that cannot be broken into smaller pieces. Labeled with lowercase letters
- *Processor*: the unit (such as a person) that completes a task. Labeled as P_1, P_2, etc. At any time, a processor is either idle or busy performing a task.
- At any stage of a project, a task can be in one of four states:
  - *eligible*: the task can be performed
  - *ineligible*: the task cannot be performed
  - *in execution*: the task is currently being performed
  - *completed*: the task has been completed

  A task is eligible when all tasks on which it depends have been completed.

Definitions

Definition

Consider a project containing multiple parts of steps.

- *Processing time* of a task: the time it takes to complete the task, denoted by $\mathrm{pt}(v)$ for task v
- *Precedence relationship*: task b relies on the completion of task a before it can be eligible
- *Finishing time* of a schedule is the total time used in that schedule
- *Optimal time* of a project is the minimum finishing time among all possible schedules, denoted OPT.

Tasks and schedules – example

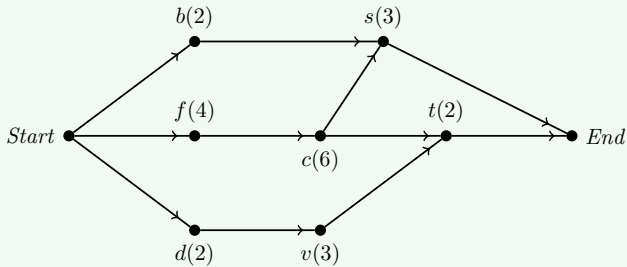
Example

- Party planning

Task	Vertex Name	Processing Time	Precedence Relationships
Buy Food	f	40	
Buy Drinks	b	20	
Dust	d	20	
Vacuum	v	30	d
Cook Food	c	60	f
Set Out Drinks	s	30	b, c
Set Table	t	20	v, c

Tasks and schedules – example

Example



- It is customary to include vertices representing the start and end of the project, and to place vertices so as to avoid edge crossings whenever possible.
- The processing times are shown in parentheses next to the vertex labels
- Edges: precedence relationships

Priority List Model

- Once a digraph has been created, the next step is to determine which processor (or person) should complete each task
- This may be easy for a project with only a few tasks, or when the dependencies between tasks are not complex.
- As complexity grows, we will need a procedure for assigning tasks
- *Priority List Model*:
  - Whenever a processor becomes available, assign the highest-priority eligible task that has not yet been assigned.
  - Precedence relationships (displayed in the digraph) determine eligibility. Ineligible tasks are skipped until all their predecessors have been completed.

Priority List Model – example

Example

- Continuing from the previous example
- Take priority list: $b - d - t - v - s - f - c$
- Consider two processors
- Each step represents a moment in time where a decision must be made

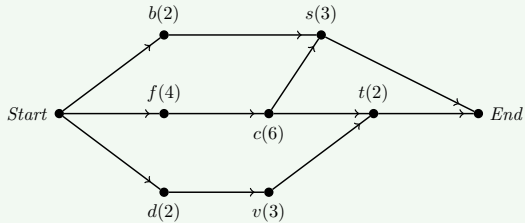
	10	20	30	40	50	60	70	80	90	100	110	120	130	140	150
P_1	b	b													
P_2	d	d													

- Step 1. $T = 0$ The first item in the priority list is b, since b does not rely on any other task, assign it to P_1. The next item d is also eligible. Assign d to P_2.

Priority List Model – example

Example

Priority list: $b - d - t - v - s - f - c$



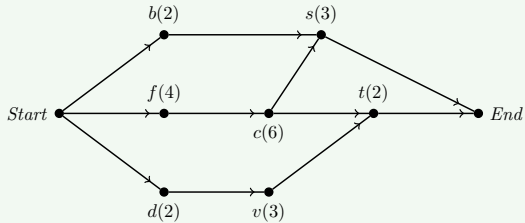
	10	20	30	40	50	60	70	80	90	100	110	120	130	140	150
P_1	b	b	v	v	v										
P_2	d	d	f	f	f	f									

- Step 2. $T = 20$. The next point at which a processor is free to pick up a task is at 20 minutes. The next task on the list is t, but it is ineligible. Both v and f are eligible.

Priority List Model – example

Example

Priority list: $b - d - t - v - s - f - c$



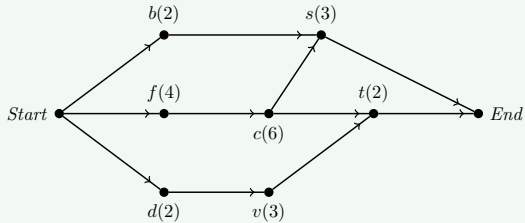
	10	20	30	40	50	60	70	80	90	100	110	120	130	140	150
P_1	b	b	v	v	v	$*$									
P_2	d	d	f	f	f	f									

- Step 3. $T = 50$. At 50 minutes, Processor 1 is ready for a new task. All tasks remaining require f to be complete. P_1 will remain idle until f is complete

Priority List Model – example

Example

Priority list: $b - d - t - v - s - f - c$



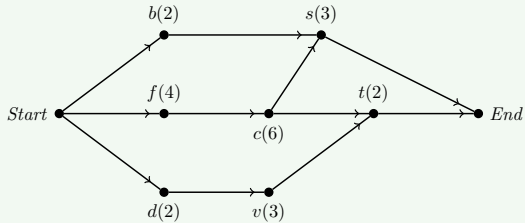
	10	20	30	40	50	60	70	80	90	100	110	120	130	140	150
P_1	b	b	v	v	v	$*$	c	c	c	c	c	c			
P_2	d	d	f	f	f	f	$*$	$*$	$*$	$*$	$*$	$*$			

- Step 4. $T = 60$. At 60 minutes, both processors are ready to take up a new task. The only eligible task is c. By convention, we assign the task to the lower indexed processor. The other processor remains idle

Priority List Model – example

Example

Priority list: $b - d - t - v - s - f - c$



	10	20	30	40	50	60	70	80	90	100	110	120	130	140	150
P_1	b	b	v	v	v	$*$	c	c	c	c	c	c	t	t	$*$
P_2	d	d	f	f	f	f	$*$	$*$	$*$	$*$	$*$	$*$	s	s	s

- Step 5. $T = 120$. Once c is complete, we can assign t to P_1 and s to P_2

The priority list $b - d - t - v - s - f - c$ yields a finishing time of 150 minutes using two processors.

Remarks

- The schedule we obtained contains a large amount of idle time, 80 minutes in total
- Although some idle time may be unavoidable, its presence should indicate that more investigation is warranted.
- The priority list given did not seem to have any connection with the digraph (in fact, it was generated randomly)
- A better approach would be to use information from the digraph to obtain a good priority list.

Shortest Paths, Project Scheduling, and Networks

- Dijkstra's Algorithm
- Chinese Postman Problem
- Project scheduling
- Critical path
- Trees
- Spanning trees
- Shortest networks
- Steiner Trees
- Metric Traveling Salesman Problem

Critical path

- *Critical path*: a path from *Start* to *End* with the greatest total processing time.
- This path is of interest because it easily identifies restrictions on the completion time of a project
- In addition, it indicates which tasks should be prioritized.
- To find the critical path, we first need to find the *critical times* of all vertices in the graph.

Critical time

Definition

The *critical time* $\mathrm{ct}[x]$ of a vertex x is the processing time of x plus the maximum critical time among all out-neighbors y of x, with $\mathrm{ct}[\mathit{End}] = 0$. For any vertex $x \neq \mathit{End}$,

$$\mathrm{ct}[x] = \mathrm{pt}(x) + \max\{\, \mathrm{ct}[y] \mid xy \text{ is an arc} \,\}$$

Note

In the definition, y is an out-neighbor of x

Critical Path Algorithm

Input: A directed acyclic project graph G with processing times given, where every task lies on a directed path from *Start* to *End*

Steps:

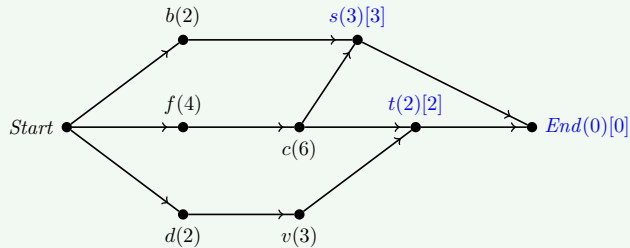
1. Label the vertex *End* with $\mathrm{pt}(\textit{End}) = 0$ and $\mathrm{ct}[\textit{End}] = 0$. For any vertex x with an arc to *End*, define $\mathrm{ct}[x] = \mathrm{pt}(x)$.
2. Process the vertices in reverse topological order, starting from *End*. Calculate the critical time of each vertex only after the critical times of all its out-neighbors have been determined.
3. Once all critical times have been obtained, find a path from *Start* to *End* by always following, from the current vertex, an outgoing arc to a neighbor of largest critical time.
4. Create a priority list by ordering vertices by decreasing critical time

Output: Critical path and critical path priority list

Critical Path Algorithm – example

Example

- Continuing from the previous example
- We will use brackets for the critical times, distinguishing them from the processing times

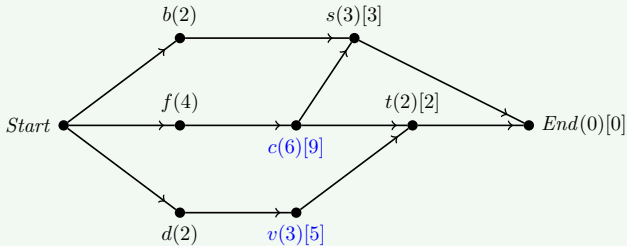


- Step 1. We express processing times in units of 10 minutes. Label *End* with critical time 0. Since s and t have arcs to *End*, set

$$\mathrm{ct}[s] = \mathrm{pt}(s) = 3, \quad \mathrm{ct}[t] = \mathrm{pt}(t) = 2$$

Critical Path Algorithm – example

Example



- Step 2. As v has a single arc to t

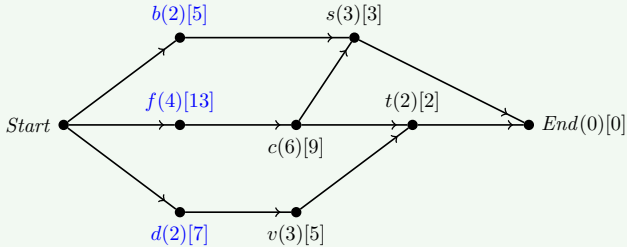
$$\mathrm{ct}[v] = \mathrm{pt}(v) + \mathrm{ct}[t] = 3 + 2 = 5$$

c has an arc to both s and t. $\mathrm{ct}[s] > \mathrm{ct}[t]$

$$\mathrm{ct}[c] = \mathrm{pt}(c) + \mathrm{ct}[s] = 6 + 3 = 9$$

Critical Path Algorithm – example

Example

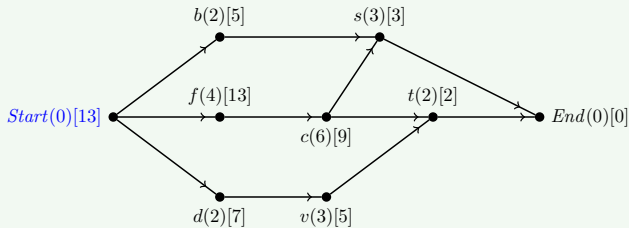


- Step 3. The remaining vertices each have a single arc to previously considered vertices

$$\mathrm{ct}[b] = \mathrm{pt}(b) + \mathrm{ct}[s] = 2 + 3 = 5$$
$$\mathrm{ct}[f] = \mathrm{pt}(f) + \mathrm{ct}[c] = 4 + 9 = 13$$
$$\mathrm{ct}[d] = \mathrm{pt}(d) + \mathrm{ct}[v] = 2 + 5 = 7$$

Critical Path Algorithm – example

Example

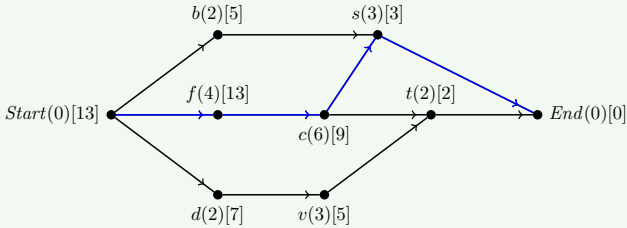


- Step 4. Label the processing time of *Start* as 0. f is the out-neighbor with the largest critical time

$$\mathrm{ct}[\mathit{Start}] = 0 + 13 = 13$$

Critical Path Algorithm – example

Example



- Step 5. Follow the path from *Start* to *End* where the vertices are chosen based on the largest critical times. This gives the path

$$Start \to f \to c \to s \to End$$

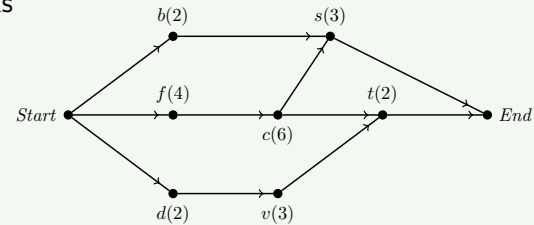
with total time 130. Ordering the vertices by decreasing critical time gives the critical path priority list

$$f - c - d - b - v - s - t$$

Priority List Model for project scheduling – example

Example

Now we use the critical path priority list $f - c - d - b - v - s - t$ to find a new schedule for the tasks



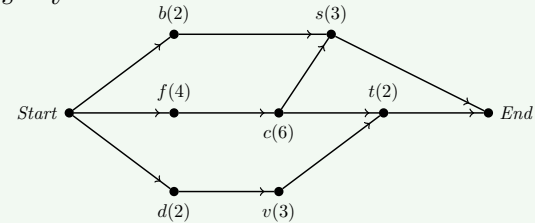
	10	20	30	40	50	60	70	80	90	100	110	120	130
P_1	f	f	f	f									
P_2	d	d											

- Step 1. $T = 0$. Since f is the first item in the list, assign it to P_1. c is not eligible. Assign d to P_2

Priority List Model for project scheduling – example

Example

$f - c - d - b - v - s - t$



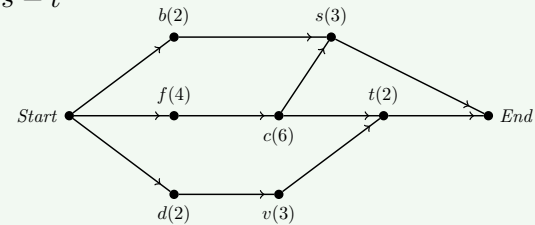
	10	20	30	40	50	60	70	80	90	100	110	120	130
P_1	f	f	f	f									
P_2	d	d	b	b									

- Step 2. $T = 20$. P_2 can be assigned a new task. Assign b, the next eligible task to P_2

Priority List Model for project scheduling – example

Example

$f - c - d - b - v - s - t$



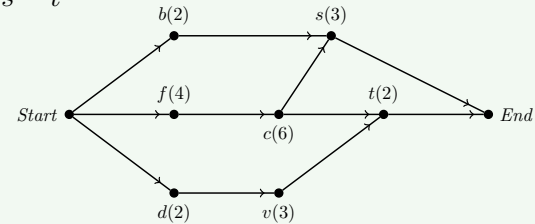
	10	20	30	40	50	60	70	80	90	100	110	120	130
P_1	f	f	f	f	c	c	c	c	c	c			
P_2	d	d	b	b	v	v	v						

- Step 3. $T = 40$. c is eligible. The next eligible task is v

Priority List Model for project scheduling – example

Example

$f - c - d - b - v - s - t$



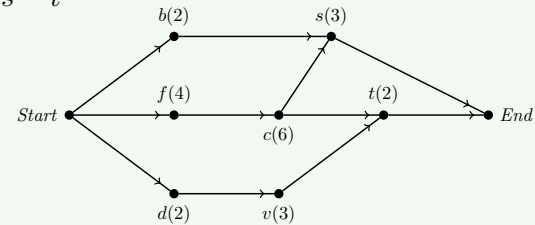
	10	20	30	40	50	60	70	80	90	100	110	120	130
P_1	f	f	f	f	c	c	c	c	c	c			
P_2	d	d	b	b	v	v	v	$*$	$*$	$*$			

- Step 4. $T = 70$. All remaining tasks are ineligible since they rely on the completion of c. P_2 remains idle.

Priority List Model for project scheduling – example

Example

$f - c - d - b - v - s - t$



	10	20	30	40	50	60	70	80	90	100	110	120	130
P_1	f	f	f	f	c	c	c	c	c	c	s	s	s
P_2	d	d	b	b	v	v	v	$*$	$*$	$*$	t	t	$*$

- Step 5. $T = 100$. s and t are now eligible
- Finishing time: 130 minutes, 40 minutes of idle time

Remarks

- Both schedules contained some idle time, though the one using the critical path priority list had only half as much as the initial example. This is partly because tasks on the critical path were prioritized.
- The schedule above must be optimal since its finishing time is equal to the critical time of Start
- In general, the critical path priority list results in a very good, though not always optimal, schedule

Optimal schedule

- The optimal time of a schedule is no less than the critical time of *Start*

$$OPT \geq \mathrm{ct}[\textit{Start}]$$

- Calculate the sum of all processing times of all tasks. The optimal time is no less than this sum divided by the total number of processors used

$$OPT \geq \frac{\sum_v \mathrm{pt}(v)}{\text{number of processors}}$$

Optimal schedule – example

Example

With our running example, sum of all processing times is 220 minutes

- Using two processors $OPT \geq \frac{220}{2} = 110$
- Using three processors $OPT \geq \frac{220}{3} \approx 73.33$
- $\mathrm{ct}[\mathit{Start}] = 130$. Neither of the above two calculations provides additional insight into the optimal schedule.

Remark

The schedule constructed earlier shows that two processors are sufficient to achieve the optimal finishing time of 130 minutes, since this equals the critical-path lower bound. Consequently, additional processors cannot reduce the optimal finishing time.

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Definition

Definition

A graph G is

- *acyclic* if there are no cycles or circuits in the graph
- a *network* if it is connected
- a *tree* if it is an acyclic network, i.e. acyclic and connected
- a *forest* if it is an acyclic graph

A vertex of degree 1 is called a *leaf*.

Trees – example

Example

Game

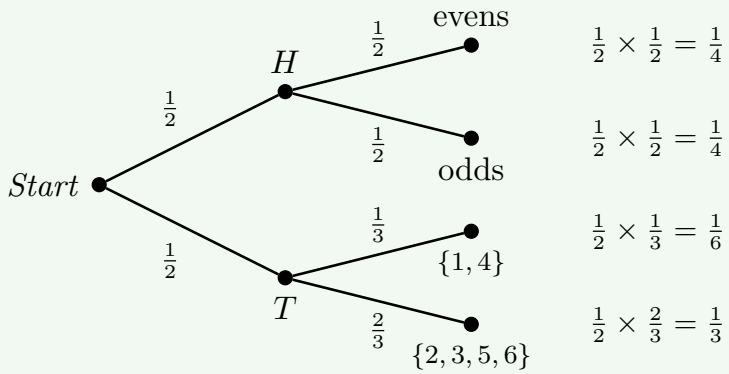
- Adam flips a coin, Jano rolls a die
- Adam gets heads, Jano rolls an even number $\rightarrow$ Jano wins 2 Euros
- Adam gets heads, Jano rolls an odd number $\rightarrow$ Adam wins 3 Euros
- Adam gets tails, Jano rolls 1 or 4 $\rightarrow$ Jano wins 5 Euros
- Adam gets tails, Jano rolls $2, 3, 5$, or 6 $\rightarrow$ Adam wins 2 Euros

A probability tree – vertices representing possible outcomes, edges labeled with the probability of the outcome

Trees – example

Example

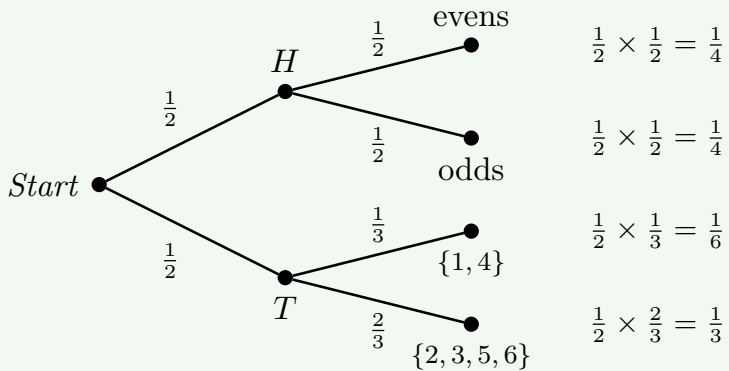
- Adam gets heads, Jano rolls an even number $\rightarrow$ Jano wins 2 Euros
- Adam gets heads, Jano rolls an odd number $\rightarrow$ Adam wins 3 Euros
- Adam gets tails, Jano rolls 1 or 4 $\rightarrow$ Jano wins 5 Euros
- Adam gets tails, Jano rolls $2, 3, 5$, or 6 $\rightarrow$ Adam wins 2 Euros



Trees – example

Example

- The probability Jano wins 5 Euros (tails and 1 or 4) is $\frac{1}{6}$
- The probability that Jano wins any money is $\frac{1}{6} + \frac{1}{4} = \frac{5}{12}$



Trees – example

Example

- Trees can be used to store information for quick access
- Consider the following sequence of numbers

$$4,\ 2,\ 7,\ 10,\ 1,\ 3, 5$$

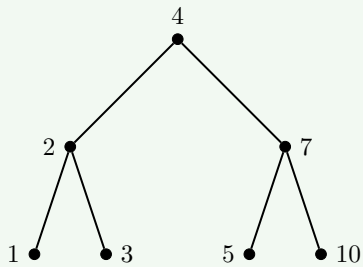
- We can form a tree by creating a vertex for each number in the list
- To insert each new number into the binary search tree, start at the root. Compare it with the current vertex and move left if it is smaller or right if it is greater. Repeat until an empty position is reached, and insert the new number there.
- If we add the restriction that no vertex can have more than two edges coming down from it, then we are forming a binary tree.

Trees – example

Example

- To insert each new number into the binary search tree, start at the root. Compare it with the current vertex and move left if it is smaller or right if it is greater. Repeat until an empty position is reached, and insert the new number there.
- Each vertex has at most two children.

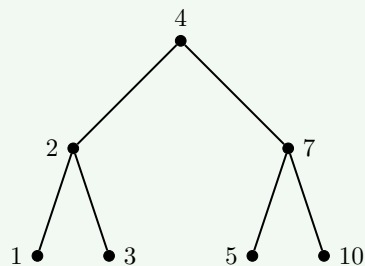
$$4,\ 2,\ 7,\ 10,\ 1,\ 3,\ 5$$



Trees – example

Example

- If we want to search for an item, we may need far fewer comparisons than in a linear search.
- Find item 5: first compare it to 4 and move right, then compare it to 7 and move left, and finally compare it to 5.
- Only three comparisons are needed to find the item.



Properties of trees

1. For every $n \geq 1$, any tree with n vertices has $n - 1$ edges
2. For any tree with $n \geq 1$ vertices, the sum of the degrees is $2n - 2$
3. Every tree with at least two vertices contains at least two leaves
4. Any network on n vertices with $n - 1$ edges must be a tree
5. For any two vertices in a tree, there is a unique path between them
6. The removal of any edge of a tree will disconnect the graph

Note

- $1 \Rightarrow 2$
- By counting the degrees of vertices, $2 \Rightarrow 3$

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- Spanning trees
- Shortest networks
- Steiner Trees
- Metric Traveling Salesman Problem

Spanning Tree

Definition

- A *subgraph* H of a graph G is a graph such that $V(H) \subseteq V(G)$ and $E(H) \subseteq E(G)$.
- H is a *spanning subgraph* if $V(H) = V(G)$
- A *spanning tree* is a spanning subgraph that is also a tree.

Note

- If an edge appears in a subgraph, then both endpoints must also be included in the subgraph
- If a vertex appears in a subgraph, any number of its incident edges may be included

Kruskal's Algorithm

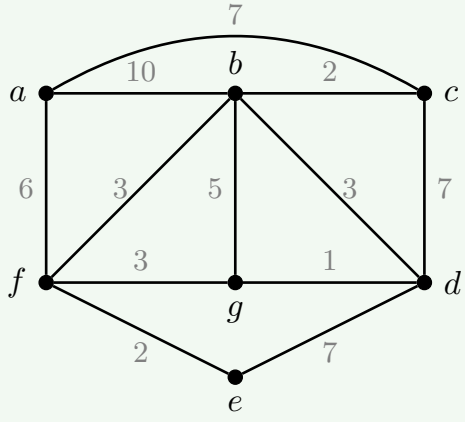
- First published in 1956 by Joseph Kruskal, an American mathematician
- Input: weighted connected graph $G = (V, E, \omega)$
- Steps
  1. Among all unhighlighted edges whose addition would not create a cycle, choose one of minimum weight. Highlight it and add it to $T = (V, E', \omega')$.
  2. Repeat Step 1 until T contains $|V| - 1$ edges.
- Output: minimum spanning tree (MST) T of G

Note

- If several eligible edges have the same minimum weight, choose one arbitrarily.
- At each step of the algorithm we are building a forest subgraph that will eventually result in a spanning tree

Kruskal's Algorithm – example

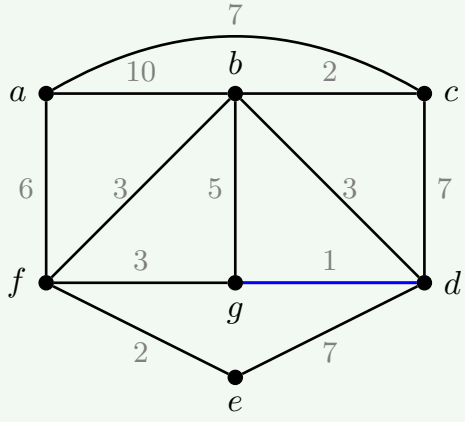
Example



- Step 1. edge with the least weight: gd

Kruskal's Algorithm – example

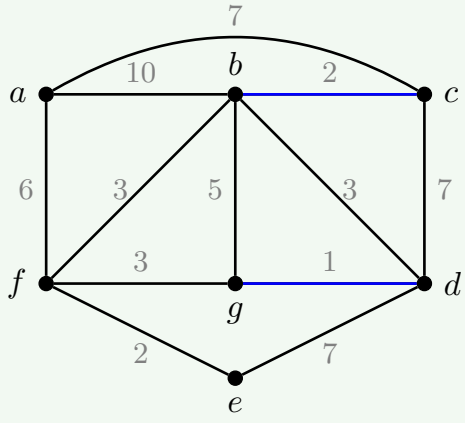
Example



- Step 1. two choices: ef and bc, let's choose bc

Kruskal's Algorithm – example

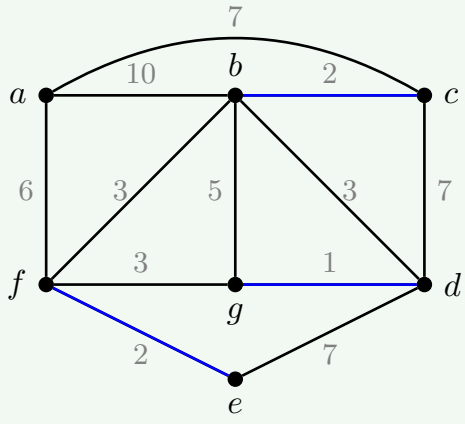
Example



- Step 1. the other edge with weight 2 is still a valid choice. Highlight ef

Kruskal's Algorithm – example

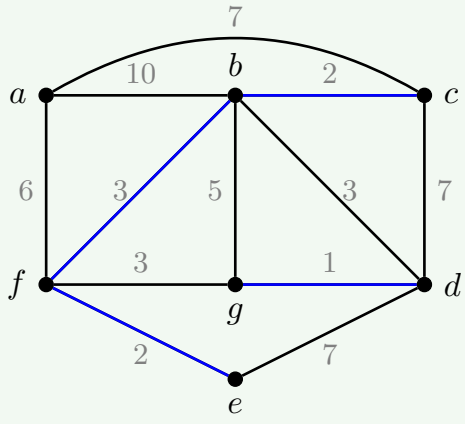
Example



- Step 1. the next smallest edge weight is 3. We randomly pick bf

Kruskal's Algorithm – example

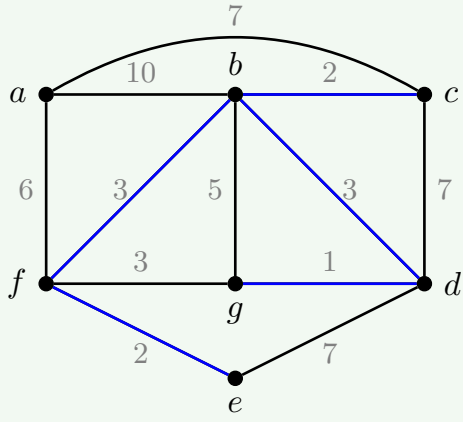
Example



- Step 1. both of the other edges of weight 3 are still available. We choose bd

Kruskal's Algorithm – example

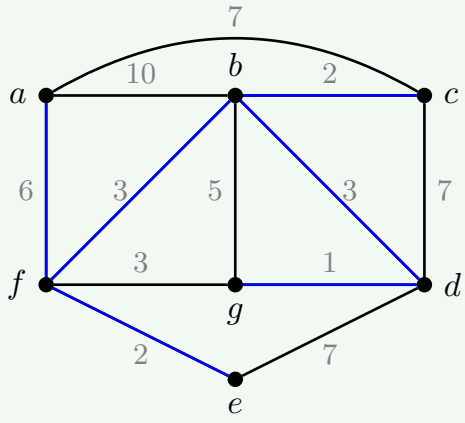
Example



- Step 1. cannot choose fg. The next smallest edge weight is 5. But we cannot choose bg. Next available edge is af of weight 6

Kruskal's Algorithm – example

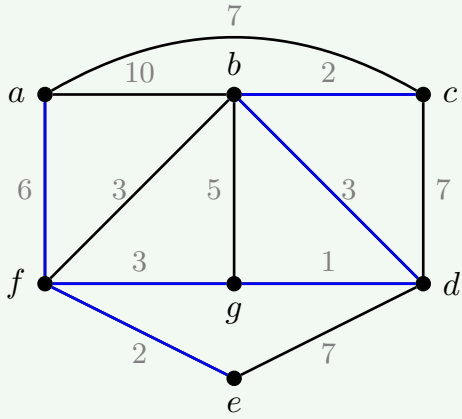
Example

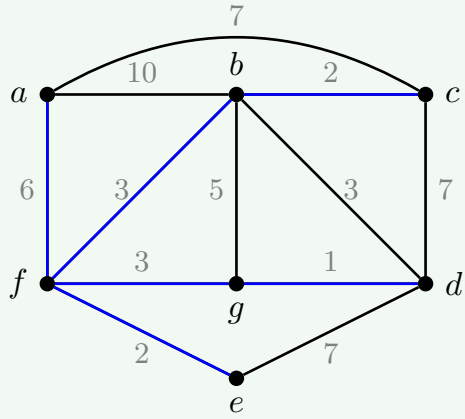


- Step 2. We now have a connected graph containing all 7 vertices and 6 edges, with no cycles. Hence, it is a tree, and we can stop. The MST has total weight 17.

Kruskal's Algorithm – example

Example





- When we were choosing an edge of weight 3, we could have made a different choice.

Kruskal's Algorithm – remarks

- When we skipped over an edge, e.g. bg of weight 5, we did so because including it would create a cycle
- This means a path between the endpoints of that edge, e.g. b and g, must already exist and the other edges along that path must each be of weight no greater than the edge we skip over
- When finding a minimum spanning tree, an edge that is strictly heavier than every other edge in a cycle cannot belong to any minimum spanning tree. If several edges share the maximum weight, some of them may belong to a minimum spanning tree.

Prim's Algorithm

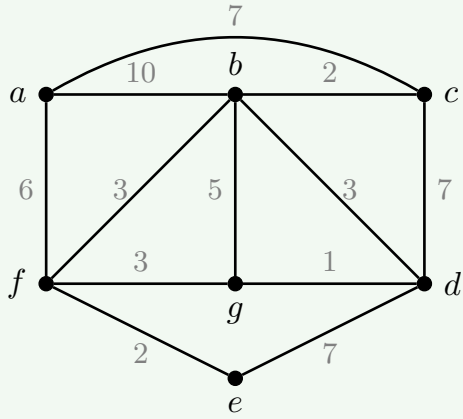
- Named after Robert C. Prim, American mathematician and computer scientist, published in 1957
- Originally discovered by Vojtěch Jarnik, Czech mathematician, in 1930
- Root: a clear starting point for the tree

Prim's Algorithm

- Input: Weighted connected graph $G = (V, E)$
- Steps
  1. Let v be the root. If no root is specified, choose a vertex at random. Highlight it and add it to $T = (V', E')$
  2. Among all edges incident to v, choose the one of minimum weight. Highlight it. Add the edge and its other endpoint to T.
  3. Let S be the set of all edges with exactly one endpoint in $V(T)$. Choose an edge of minimum weight from S, highlight it, and add this edge and its endpoint outside $V(T)$ to T.
  4. Repeat Step 3 until T contains all vertices of G.
- Output: rooted MST T of G

Prim's Algorithm – example

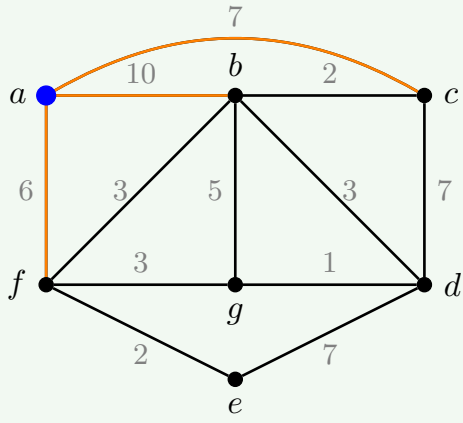
Example



- Same example as before
- Step 1. Choose a as the starting vertex

Prim's Algorithm – example

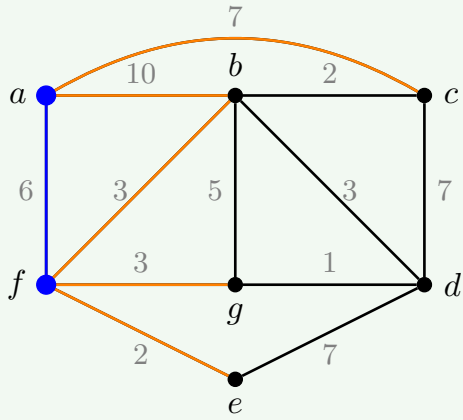
Example



- Step 2. among the edges incident to a, af, ab, ac, the edge of the least weight is af

Prim's Algorithm – example

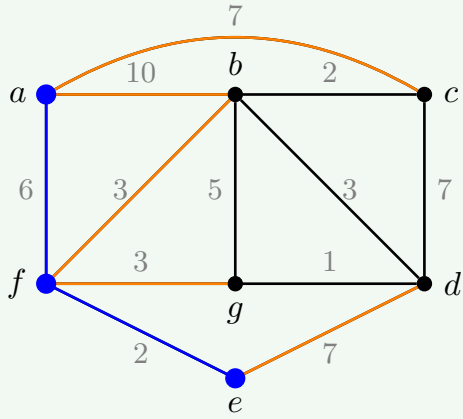
Example



- Step 3. S consists of edges with exactly one endpoint in the current tree, whose vertices are a and f. The edge with the minimum weight is ef

Prim's Algorithm – example

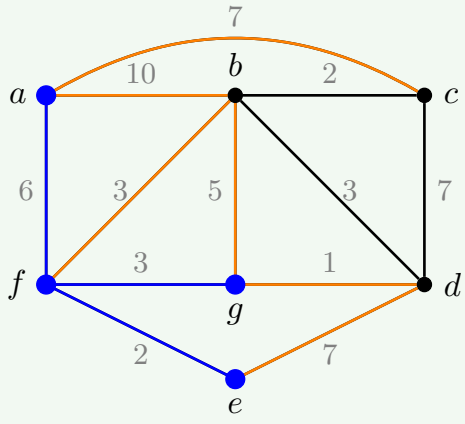
Example



- Step 3. S consists of edges with exactly one endpoint in the current tree, whose vertices are a, e, and f. The minimum-weight edges are fb and fg. Let us choose

Prim's Algorithm – example

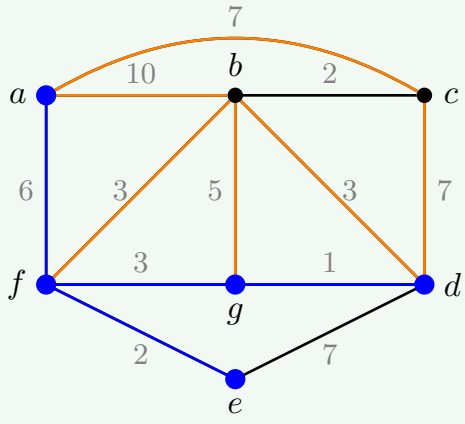
Example



- Step 3. The next edge to add to the tree is dg

Prim's Algorithm – example

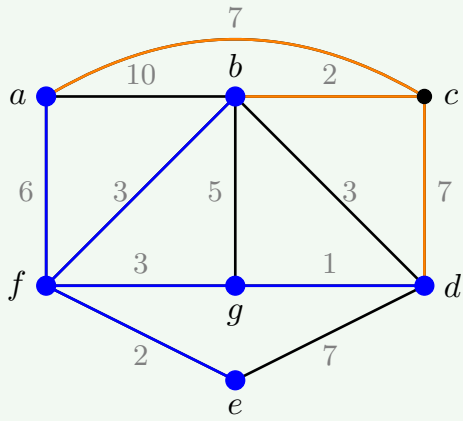
Example



- Step 3. There are two possible minimum weight edges, bf or bd. We choose bf

Prim's Algorithm – example

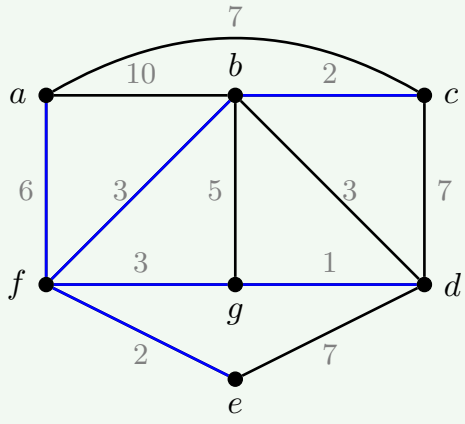
Example



- Step 3. Now the only edges to consider are those with one endpoint of c since this is the only vertex not part of our tree. The edge of minimum weight is bc

Prim's Algorithm – example

Example



- We have obtained a MST of weight 17

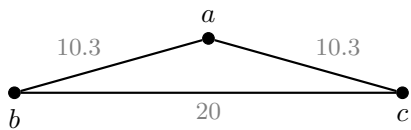
Shortest Paths, Project Scheduling, and Networks

- Dijkstra's Algorithm
- Chinese Postman Problem
- Project scheduling
- Critical path
- Trees
- Spanning trees
- Shortest networks
- Steiner Trees
- Metric Traveling Salesman Problem

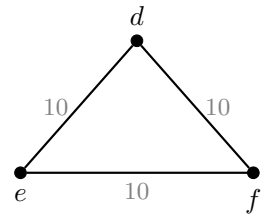
Shortest network

- Network: a connected graph
- A shortest network is a connected network of least total weight.

MST



T_1



T_2

- Edge lengths are drawn to scale since the angle created by the edges will play a large role in determining where to place shortcuts
- The minimum spanning tree consists of the two edges of length 10.3 in T_1, and any two edges in T_2.
- Could we do better than a minimum spanning tree?

Fermat point

- A similar question was posed by the 17th century French mathematician Pierre de Fermat in a letter to Evangelista Torricelli, an Italian physicist and mathematician
- In his letter, Fermat challenged Torricelli to find a point that minimizes the sum of the distances to the three vertices of a triangle.

Definition

A *Fermat point* of a triangle is a point p for which the sum of the distances from p to the three vertices is minimized. When all angles of the triangle are less than $120°$, the three angles formed at p are each $120°$.

Shortest network for a triangle

Theorem

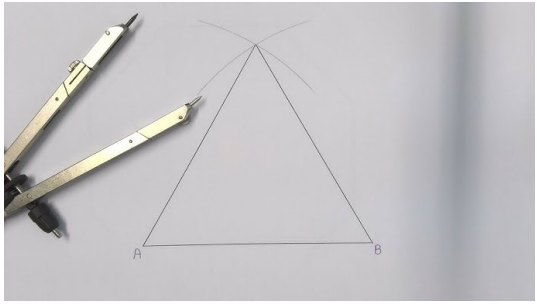
Given three points and the triangle T determined by them, the shortest network connecting the three points is either

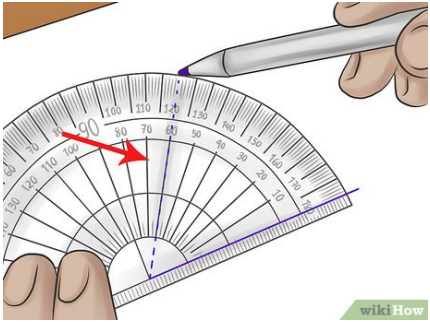
- *the two shortest sides of T provided T has one angle of at least 120°*
- *the three segments connecting the Fermat point for T to the original three vertices of T*

Torricelli's Construction

- Input: Triangle $T = \triangle abc$ where all angles measure less than 120°
- Steps
  1. Along edge ab of T construct an equilateral triangle using that edge and a new point x that is on the opposite side of the edge as c
  2. Repeat step 1 for the other two edges of T, introducing new points y and z across from b and a, respectively
  3. Join x and c, y and b, and z and a by a line segment
  4. The common intersection point of these three line segments is the Fermat point p of T.
  5. The shortest network is the line segments joining each of the original vertices, a, b, and c, with p
- Output: Fermat point p and shortest network connecting a, b, and c

Torricelli's Construction – tools



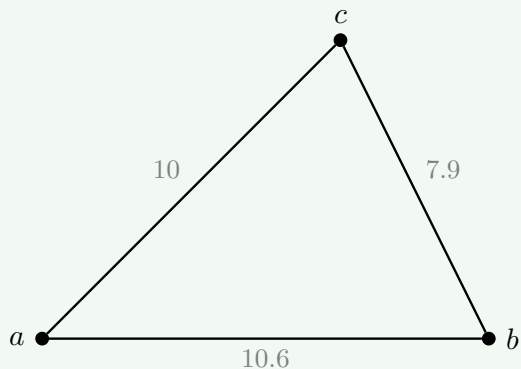


Note

Can be done using a ruler and a compass (or a protractor)

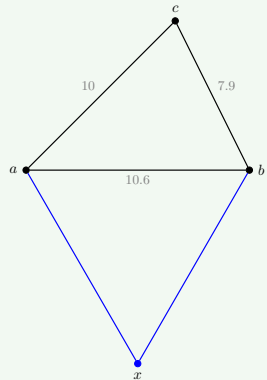
Torricelli's Construction – example

Example



Torricelli's Construction – example

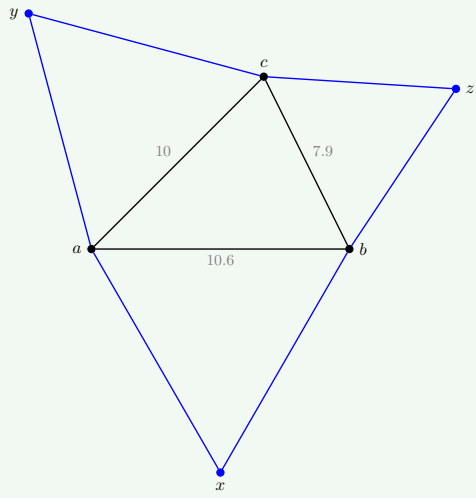
Example



- Step 1. Form an equilateral triangle off edge ab with new point x on the opposite side of ab as c

Torricelli's Construction – example

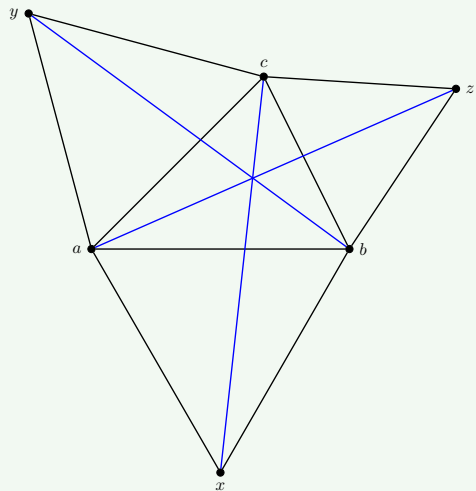
Example



- Step 2. Repeat the construction for edges ac and bc.

Torricelli's Construction – example

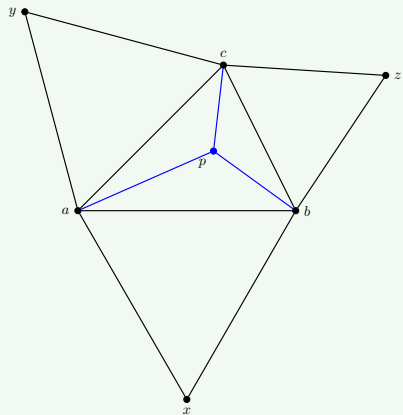
Example



- Step 3. Join x and c, y and b, and z and a

Torricelli's Construction – example

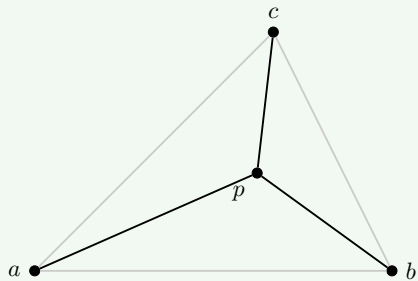
Example



- Step 4. Find the Fermat point p as the common intersection point of the three constructed segments.
- Step 5. Highlight the edges from p to each of the original vertices

Torricelli's Construction – example

Example



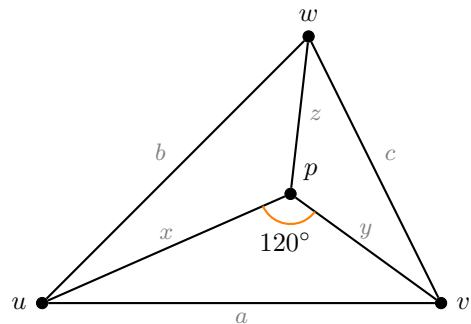
- Output: the shortest network consists of the three segments joining p to the original vertices.

Torricelli's Construction – length of the shortest network

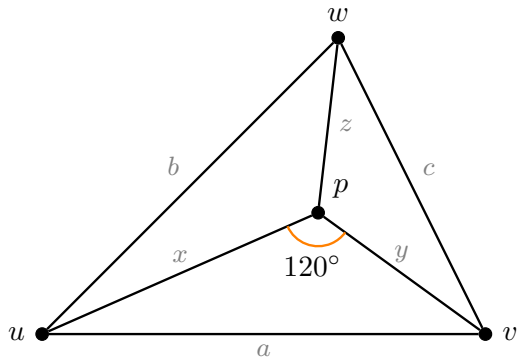
- Law of cosines

$$\begin{aligned} a^2 &= b^2 + c^2 - 2bc\cos(\angle uwv) \\ c^2 &= a^2 + b^2 - 2ab\cos(\angle wuv) \\ b^2 &= a^2 + c^2 - 2ac\cos(\angle wvu) \end{aligned}$$

- Compute the angles of the triangle
- Assign coordinates to vertices
- Solve for the lengths x, y, and z



Torricelli's Construction – length of the shortest network



- If ℓ denotes the length of the shortest network connecting vertices of a triangle having angles that measure less than 120° each and sides of lengths a, b, c

$$\ell = x + y + z = \sqrt{\frac{a^2+b^2+c^2}{2} + \frac{\sqrt{3}}{2}\sqrt{2a^2b^2+2a^2c^2+2b^2c^2-(a^4+b^4+c^4)}}$$

Torricelli's Construction – length of the shortest network

Example

- Continuing from the previous example

$$a = 7.9, \quad b = 10, \quad c = 10.6$$

- With the formula we get the length of the network obtained

$$\ell = \sqrt{\frac{274.77}{2} + \frac{\sqrt{3}}{2}\sqrt{48978.7752 - 26519.7777}} \approx 16.345$$

- In comparison, the MST has a total length of 17.9
- The shortest network saves 1.555, or 8.69%

Shortest Paths, Project Scheduling, and Networks

- Dijkstra's Algorithm
- Chinese Postman Problem
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Steiner Trees

Definition

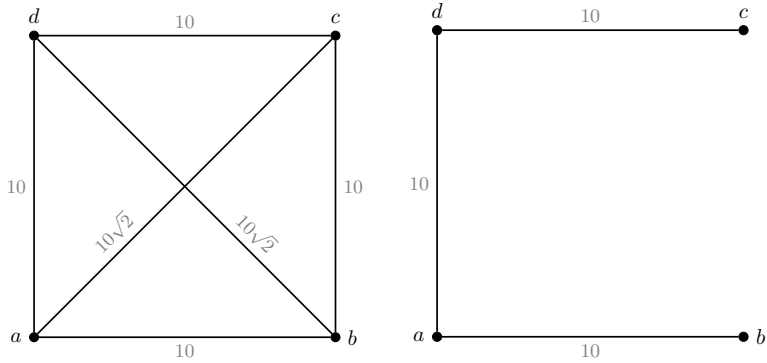
- For a graph G, a *Steiner point* is a new point p added to the graph with degree 3, such that the three edges incident to p meet at angles of 120°.
- A *Steiner tree* is a tree whose vertices consist of the original vertices of G together with possible Steiner points.

- A Steiner point is similar to a Fermat point in that it is added to a graph to help find a shortest network connecting the original vertices.
- It can be shown that finding a shortest network amounts to finding a minimum Steiner tree.
- This problem is named for the 19th century Swiss mathematician Jakob Steiner

Finding Steiner Trees

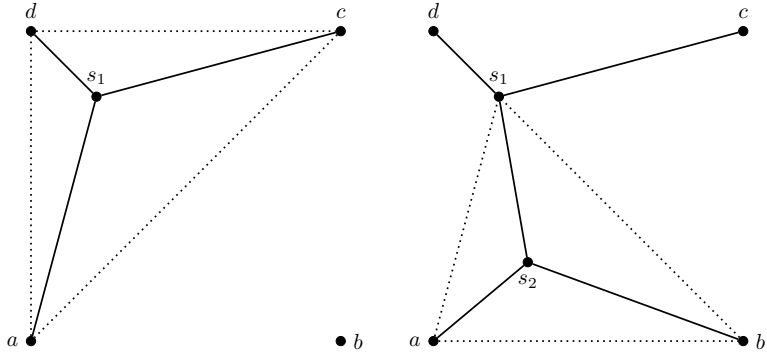
- Although the Steiner Tree Problem sounds fairly simple, it is NP-hard.
  - Finding an optimal solution can be computationally difficult.
  - No efficient algorithm is known for solving the problem in general.

Four vertices – MST



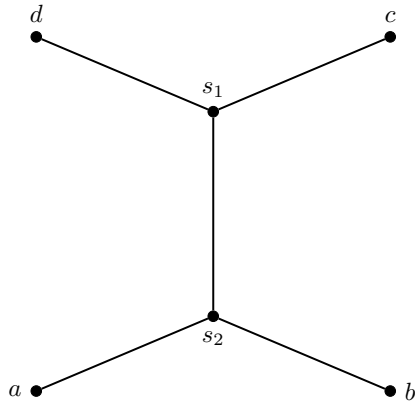
- A graph with four vertices that lie on a square
- A minimum spanning tree consists of three of the four edges of length 10.

Four vertices – Fermat points



- s_1: Fermat point for $\triangle adc$
- s_2: Fermat point for $\triangle abs_1$
- Length of network: ≈ 27.852
- These are not Steiner points, since the incident edges do not form 120° angles.

Four vertices – Steiner tree



- Total length 27.321
- Network with Fermat points: 27.852
- MST: 30

Steiner Network Method

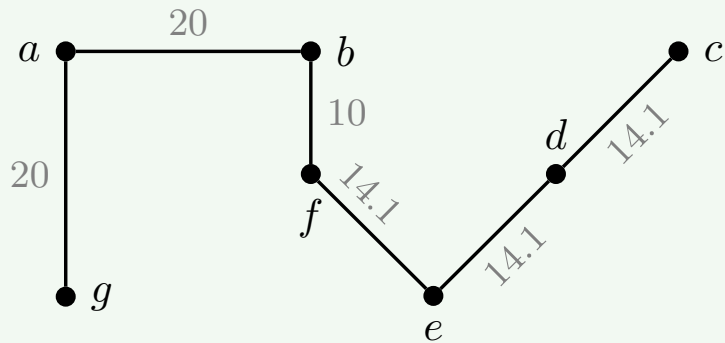
- We will not attempt to find a minimum Steiner tree.
- Instead, we use the ideas behind Steiner trees to find a network shorter than a minimum spanning tree, if possible.

Steiner Network Method

- Step 1. Find the MST of the network
- Step 2. Form a triangle using two edges of the current network that share a vertex. If all angles of this triangle are less than 120°, find its Fermat point.
- Step 3. Update the network by removing the two edges used in Step 2 and adding edges from the corresponding vertices to the Fermat point.
- Step 4. Repeat steps 2 and 3 until all possible triangles have been considered

Steiner Network Method – example

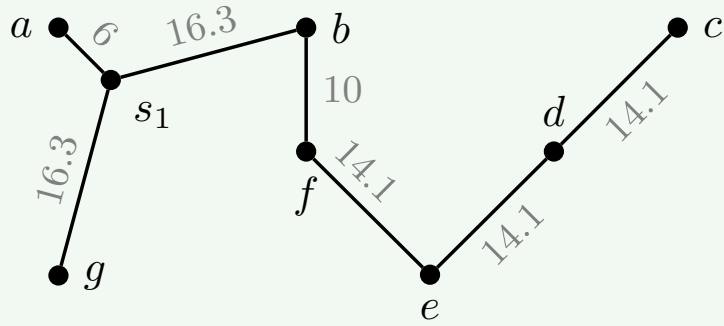
Example



- We begin with vertices a, b, g, since the angle at a is $90°$.

Steiner Network Method – example

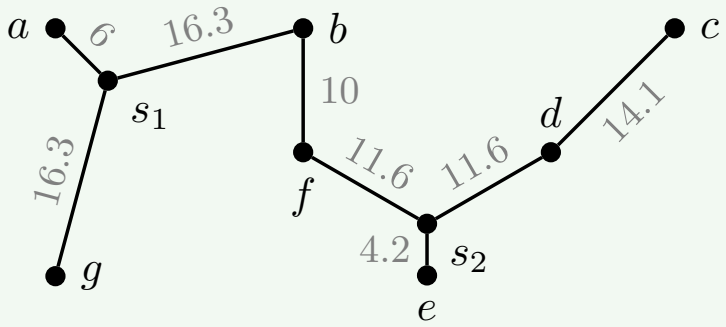
Example



- We begin with vertices a, b, g. Using Torricelli's Construction, we obtain the point s_1.

Steiner Network Method – example

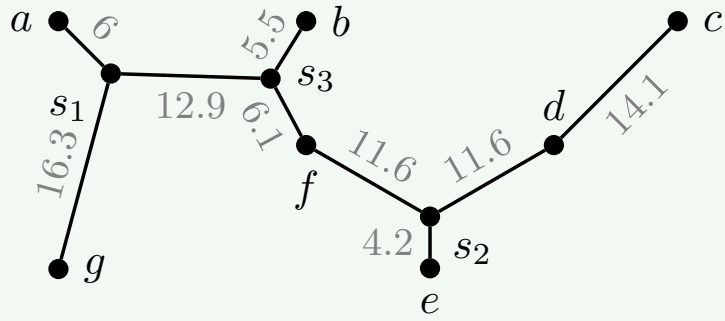
Example



- Next, consider vertices f, e, d. Since the angle at e is $90°$, Torricelli's Construction gives the point s_2.
- At this stage, the angle conditions leave only one triangle to consider: b, f, s_1.

Steiner Network Method – example

Example



- Finally, using vertices b, f, s_1, we obtain the point s_3.
- Total length: 88.3
- Original (MST) length: 92.3

Remark

- Although we have not fully solved the optimization problem for networks containing more than three points,
- the Steiner Network Method provides a quick and simple way to identify possible improvements to a minimum spanning tree.

Shortest Paths, Project Scheduling, and Networks

- Dijkstra's Algorithm
- Chinese Postman Problem
- Project scheduling
- Critical path
- Trees
- Spanning trees
- Shortest networks
- Steiner Trees
- Metric Traveling Salesman Problem

Traveling Salesman Problem

- How should a delivery service plan its route through a city to ensure each customer is reached?
- A traveling salesman has customers in numerous cities. He must visit each of them and return home, but wishes to do this with the least total cost
- The Traveling Salesman Problem asks for a shortest cycle that visits every city exactly once and returns to the starting city; in graph terms, this is a shortest Hamiltonian cycle in a weighted complete graph.

Metric Traveling Salesman Problem

- Abbreviated as mTSP
- Only considers the scenarios where the weights satisfy the *triangle inequality*
- For a weighted complete graph $K = (V, E, \omega)$, given any three vertices x, y, z

$$\omega(xy) + \omega(yz) \geq \omega(xz)$$

mTSP Algorithm

- mTSP Algorithm combines three ideas we have studied so far – Eulerian circuits, Hamiltonian cycles and MST
- We duplicate every edge of a minimum spanning tree, making all vertex degrees even and thus producing a graph with an Eulerian circuit.
- We then convert this Eulerian circuit into a Hamiltonian cycle by skipping repeated vertices.
- This procedure is guaranteed to produce a Hamiltonian cycle only when the underlying graph is complete.
- It may still find a proper Hamiltonian cycle when the graph is not complete, but cannot be guaranteed to do so.

mTSP Algorithm

- Input: weighted complete graph K_n, where the weight function ω satisfies the triangle inequality
- Steps:
  1. Find a MST T for K_n
  2. Duplicate all the edges of T to obtain T^*
  3. Find an Eulerian circuit for T^*
  4. Convert the Eulerian circuit into a Hamiltonian cycle by skipping any previously visited vertex (except for the starting and ending vertex)
  5. Calculate the total weight
- Output: Hamiltonian cycle for K_n

mTSP Algorithm – example

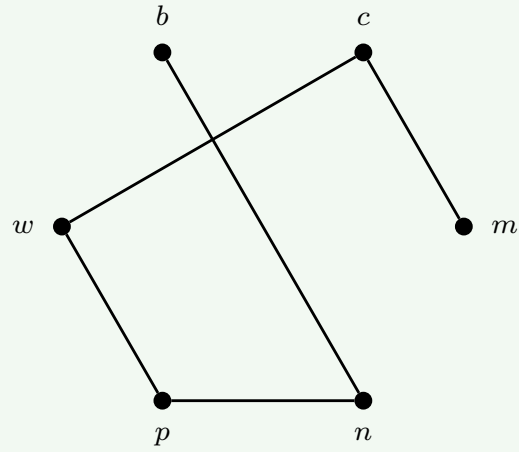
Example

Alice must visit six cities and needs to minimize the total distance

	Boston	Charlotte	Memphis	New York	Philadelphia	D.C.
Boston	.	840	1316	216	310	440
Charlotte	840	.	619	628	540	400
Memphis	1316	619	.	1096	1016	876
New York City	216	628	1096	.	97	228
Philadelphia	310	540	1016	97	.	140
Washington, D.C.	440	400	876	228	140	.

mTSP Algorithm – example

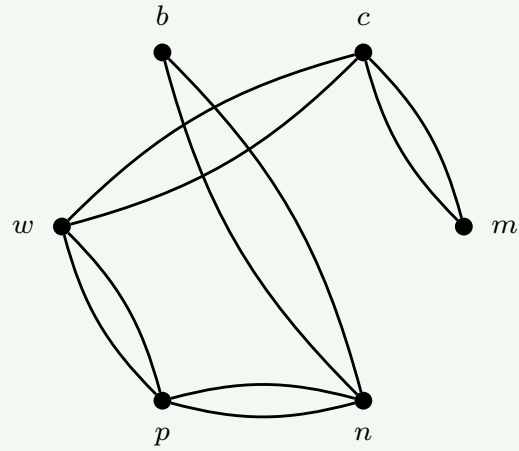
Example



- Step 1. MST

mTSP Algorithm – example

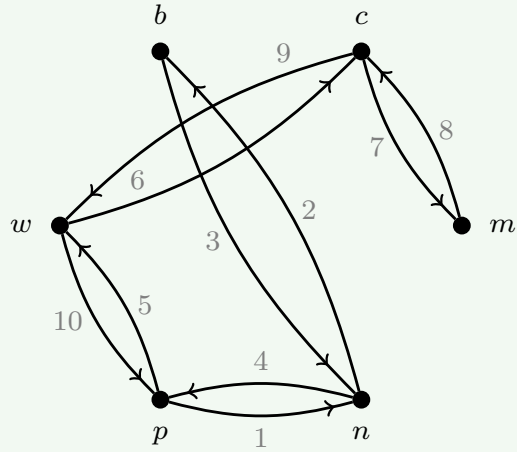
Example



- Step 2. Duplicate all edges of the MST

mTSP Algorithm – example

Example



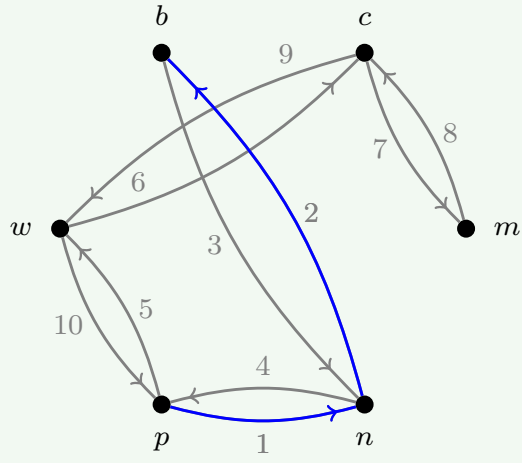
- Step 3. An Eulerian circuit is $pnbnpwcmcwp$

mTSP Algorithm – example

Example

Step 4

- Follow the Eulerian circuit until we reach b.
- Since we want a Hamiltonian cycle, we must skip any vertex that has already been visited, so we do not return to n.
- The next vertex along the circuit that has not been previously visited is w

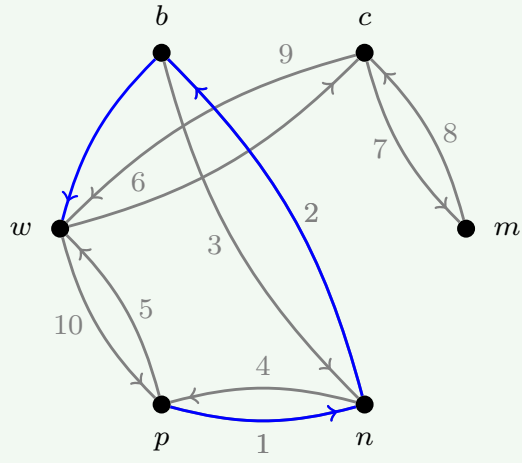


mTSP Algorithm – example

Example

Step 4

- Add the edge from b to w

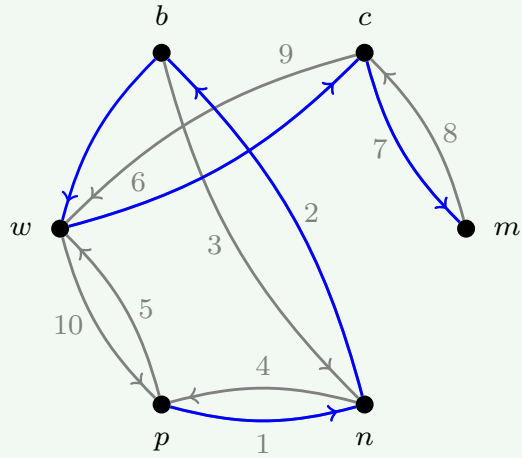


mTSP Algorithm – example

Example

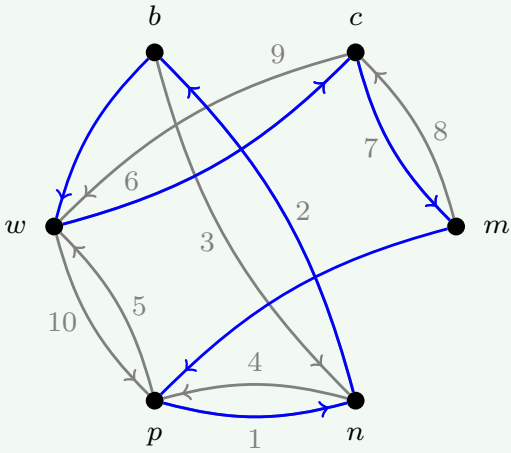
Step 4

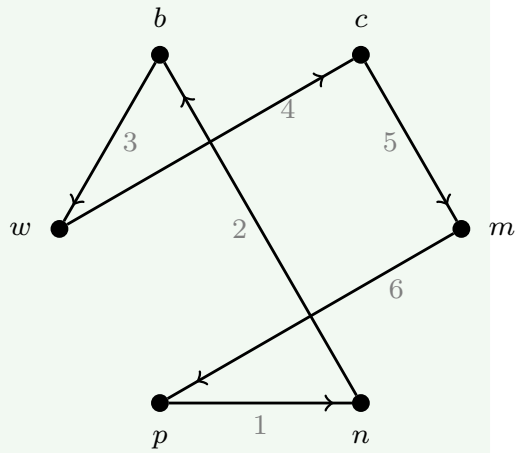
- Continue following the Eulerian circuit until m is reached.
- Again, we skip c because it has already been visited, and at this stage the only remaining move is to return to p.



mTSP Algorithm – example

Example





mTSP Algorithm – example

Example

- Total weight of the MST: 1472 – the worst possible Hamiltonian cycle that can arise from it will have weight at most two times that 2944, due to the doubling of the edges
- Total weight of the Hamiltonian cycle: 2788
- The actual optimal Hamiltonian cycle has total weight 2781.
- Thus, the relative error of our Hamiltonian cycle is approximately 0.25%.